

sum.

For IIR filter, the convolution sum is given by:

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) \dots \dots \text{Eq. (1)}$$

For FIR, the convolution sum is given by:

$$y(n) = \sum_{k=0}^{N-1} h(k)x(n-k) \dots \dots \text{Eq. (2)}$$

The IIR filtering equation is expressed in a recursive form since the length of Eq. (1) is infinitely long. It is given by:

$$y(n) = \sum_{k=0}^N b_k x(n-k) - \sum_{k=1}^M a_k y(n-k) \dots \dots \text{Eq. (3)}$$

From Equations (1) and (2) it is clear that the impulse response of IIR filters is of infinite duration, while for FIR it is of finite duration.

It is evident from Eq. (3) that IIR filter is some sort of a feedback system since current output $y(n)$ is a function of past outputs as well as present and past input samples. However, in FIR, current output sample $y(n)$ is a function only of past and present input values. Also, the values of 'h(k)' for FIR, and 'ak' and 'bk' (filter coefficients) for IIR are very important objectives of most filter design problems.

IIR filters can be used in the generation and detection of the dual-tone multi-frequency (DTMF) signal used by a touch-tone phone. The block diagram of a DTMF generator using two IIR filters connected in parallel is shown in Fig. 1. Besides the use as dialing tone, the DTMF signal can also be used for selecting options in telephone servicing systems.

MATLAB programs

IIR filter. Install MATLAB version 7 in your system and enter the `iir.m` code given at the end of this article. The `iir.m` MATLAB code is written as per the Eq. (1) given above. Compile and run the code. The program plots the magnitude and phase responses as per the 3dB frequency entered by the user as shown in Fig. 2.

FIR filter. Two very different design techniques are commonly used for digital FIR filters—windows and

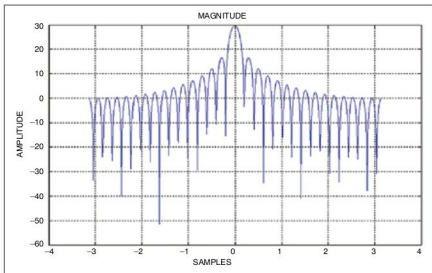


Fig. 3: Magnitude plot for rectangular filter

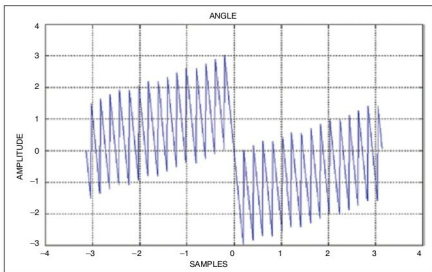


Fig. 4: Angle plot for rectangular filter

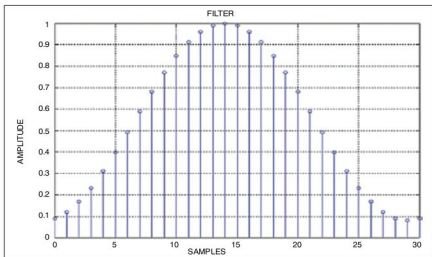


Fig. 5: Signal input to Hamming filter